

Chapter 2

Wave Functions and Observables

Lecture 3: Wave functions

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The first two lectures collected the experimental facts that force us away from classical mechanics: repeatable binary measurements, interference, diffraction, discrete spectra, the photoelectric effect, and reproducible randomness. We now begin the mathematical language of quantum mechanics.

The goal of this lecture is modest: introduce the wave function as the object that replaces the classical state (x, p) , and connect it to the de Broglie relations.

Questions.

- What replaces the classical state (x, p) ?
- Why should particles have wave-like descriptions?
- What is the simplest wave associated with a definite momentum?
- Why can position and momentum not both be perfectly sharp?

2.1 From experimental motivation to the wave function

2.1.1 A reminder from polarization

Repeatability Polarizers give a visible version of the color-hardness box experiments. Let vertical polarization play the role of White and horizontal polarization play the role of Black. A vertical polarizer prepares vertically polarized light. A second vertical polarizer transmits it, while a horizontal polarizer blocks it.

Intermediate measurement If a 45° polarizer is inserted between the vertical and horizontal polarizers, some light passes through. The middle polarizer changes the state before the final test. This is the same lesson as the box experiments: an intermediate measurement is not passive.

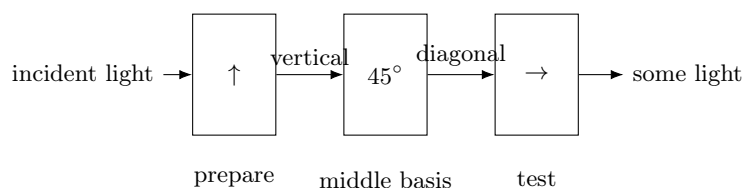


Figure 2.1: A middle polarizer between crossed polarizers makes a nonzero final intensity possible.

For ideal polarizers, Malus's law gives

$$I_{\text{out}} = I_{\text{in}} \cos^2 \theta, \quad (2.1)$$

where θ is the angle between the incoming polarization and the polarizer axis. Crossed polarizers give $I_{\text{out}} = 0$, but the sequence $0^\circ \rightarrow 45^\circ \rightarrow 90^\circ$ gives a nonzero transmitted intensity.

The lesson is that the state is not just a list of pre-existing answers; it can be changed by a measurement in a different basis.

2.1.2 De Broglie relations

Light has wave quantities: wavelength λ , frequency ν , angular frequency $\omega = 2\pi\nu$, and wave number $k = 2\pi/\lambda$. The photoelectric effect tells us that light also exchanges energy and momentum in packets:

$$E = h\nu = \hbar\omega, \quad p = \frac{h}{\lambda} = \hbar k, \quad \hbar = \frac{h}{2\pi}. \quad (2.2)$$

These formulas connect particle-like quantities (E, p) to wave-like quantities (ω, k) .

De Broglie's idea was that the same relations should hold for matter. A particle with momentum p should have an associated wavelength

$$\lambda = \frac{h}{p}, \quad p = \hbar k. \quad (2.3)$$

Similarly, a particle with energy E should have an associated angular frequency

$$E = \hbar\omega. \quad (2.4)$$

The Davisson–Germer experiment confirmed this prediction for electrons: electrons diffract from a crystal as if they have wavelength $\lambda = h/p$.

Proposition 1 (De Broglie relation). Matter has wave-like behavior. A particle with momentum p is associated with wave number k by

$$p = \hbar k.$$

For a particle with definite wave number k and angular frequency ω , the associated one-dimensional plane wave has the form

$$\psi(x, t) = Ae^{i(kx - \omega t)}. \quad (2.5)$$

In three dimensions this becomes $Ae^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$. For now we work in one spatial dimension.

2.1.3 Why classical states are not enough

In classical mechanics, the state of a single particle is completely specified by its position and momentum:

$$\text{classical state} = (x, p). \quad (2.6)$$

Knowing x and p is complete physical information in classical mechanics. Other quantities, such as energy, are then calculated from the state.

This cannot be the basic description of a quantum particle. The real world obeys an uncertainty relation of the form

$$\Delta x \Delta p \gtrsim \hbar, \quad (2.7)$$

Later we will sharpen this to the standard inequality $\Delta x \Delta p \geq \hbar/2$. For now, the point is simple: a particle cannot be assigned an arbitrarily precise position and an arbitrarily precise momentum at the same time. Therefore (x, p) cannot be the fundamental state of a quantum particle.

2.1.4 The wave function as the state

Definition 1 (Wave function, first pass). For a single quantum particle in one dimension, the state is completely specified by a complex-valued wave function

$$\psi(x). \quad (2.8)$$

For now, keep four points in mind:

1. $\psi(x)$ replaces the classical state (x, p) .
2. $\psi(x)$ is a complex number at each position x .
3. $\psi(x)$ must be single-valued and normalizable.
4. In one dimension, the dimension of ψ is $[\psi] = L^{-1/2}$.

Reasonable wave functions Not every imaginable curve is acceptable. At minimum, a wave function must be single-valued: it cannot assign two different complex numbers to the same position x . In ordinary finite-potential problems, physical wave functions are also continuous and sufficiently smooth. The exact technical conditions will become clearer as we work examples.

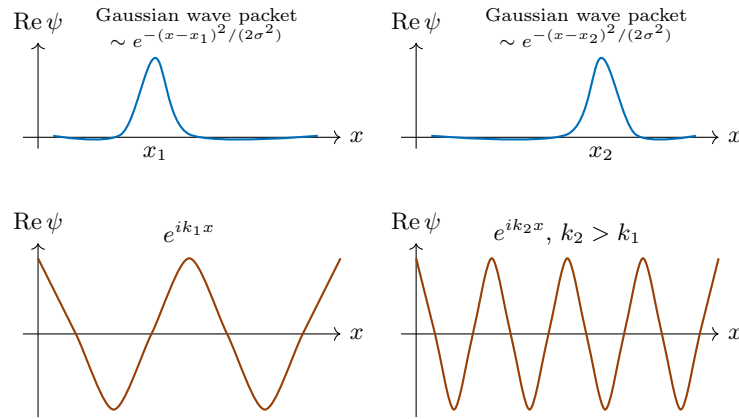


Figure 2.2: Four schematic wave functions. The top two are localized in position. The bottom two are plane waves with definite wavelength, but only the real part is shown.